STA4026S Analytics – Neural Networks.

Assignment 2 2026 (32 Marks)

"Hitting the wall." A runner's guide.

You are a Data Scientist consulting for a Must Love Hills (MLH) run crew, an informal running group. Coaches at MLH would like to know how to advise runners about running nutrition for Cape Town Marathon coming up in May 2026. The marathon distance is notorious for eliciting a response from runners of all skill levels and calibres known as "hitting the wall" or "bombing". That is, runners experience an acute drop in strength and performance (usually accompanied by pain and severe muscle fatigue) at some point during the race when their body depletes its glycogen stores. This can be mitigated by fuelling appropriately during the race. That is, taking in fluids and carbohydrates in the form of water or drinks and gels (small sachets of high concentrate carbohydrate edible gel).

The question thus remains: what is the appropriate amount of fluids and fuel to take in during the marathon in order to avoid "bombing?"

For these purposes, you have surveyed runners taking part in the Cape Town Marathon 2024 and recorded observations on fluid, carb intake, etc. and whether or not a runner hit the wall (bombed) during the race.

Variable	Description
`Bombed`	Response: Dichotomous categorical response encoded as TRUE if an athlete 'bombed', FALSE otherwise.
`AveSpeed`	Average speed over the first half of CTM, in km/h.
`Sex`	Categorical variable for sex. Value is 1 if male, 0 if female.
`Fluid`	Fluid intake (continuous) measured in litres per hour (l/h).
`Carbs`	Carbohydrate intake (continuous) measured in grams per hour (g/h).

As part of your analysis, you are tasked with writing various bits of code and subsequently making interpretive assessments through the questions below.

For purposes of modelling the present dataset, we consider a type of double hidden layer neural network, which we will term a split-brain-($2m$,2)-network, where the hidden layers are split into two **independent** sub-networks or hemispheres with an identical number of nodes (here, m) on the first hidden layer, each taking as input a distinct subset of features of the network, followed by a **single** node on each hemisphere for the **second layer**. Once evaluated, the activations from the hemispheres are treated as usual

for the remainder of the forward pass; the two activations from each of the hemispheres enter the output layer.

Figure 1 gives a directed graph of the neural network. We delineate the inputs and elements of each of the hemispheres by superscripts u and v. The example here has two inputs on each hemisphere.

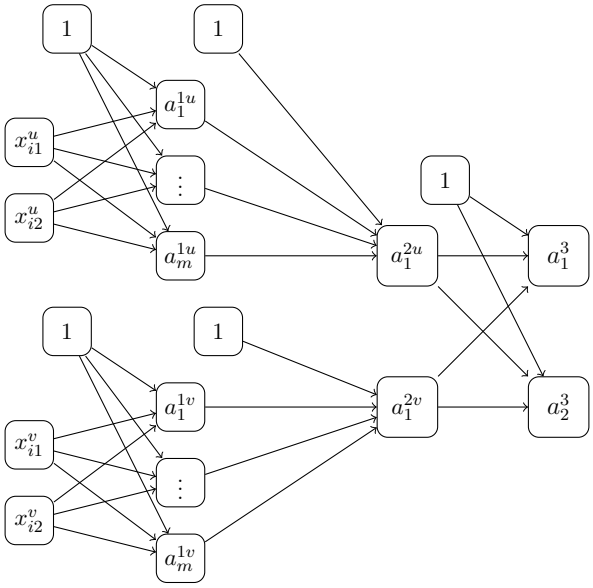

Figure 1: Directed graph of a 'split-brain' network. Note that there are two sub-networks or 'hemispheres' in the initial layers of the network, hence activations incur the additional superscripts u and v to indicate which sub-network (and inputs) they pertain.

Note: I've added leading indicators to each question telling you what should be reflected in your compiled write-up. To that end, here's what they mean explicitly:

- Render code: This means, I want to see your R code typeset at that point in your document. So make sure the code-chunk options are set such that the code renders.
- Render plots: This means, I want to see a plot rendered in the write up.
- In-text response/typeset mathematical response/written response: This means I want a paragraph-style response where you type out your response in one or a few sentences and/or typeset mathematical equations and content appropriately.
- Combinations of the above: If you see R code + plots, that means render the R code and the plot, and no typed response is required. If you see R code, that means just render the R code and no other response is required.

Problem Set

1. (a) **Render code + plot + written response**: Conduct an exploratory data analysis (EDA) by plotting input-response relationships and comment on whether non-linear machinery such as a neural network is an appropriate model class for the present problem. [3]

   (b) **Typeset mathematical and written response**: The scalar form of the softmax activation function is given by: [2]

$$\sigma(z_j) = \frac{\exp(z_j)}{\sum_{i=1}^{q} \exp(z_i)}, \tag{1}$$

   for q outputs. In practice, one actually evaluates the following:

$$\sigma(z_j) = \frac{\exp(z_j - z^\star)}{\sum_{i=1}^{q} \exp(z_i - z^\star)}, \tag{2}$$

   where $z^\star = \max\{z_1, z_2, \ldots, z_q\}$. Show that Equation 1 is equivalent to Equation 2 and explain why one would rather evaluate Equation 2.

   (c) **Render code**: Write an R-function that evaluates Equation 2 for matrix arguments. That is, if we let $\mathbf{Z}^L = [\mathbf{z}^L(1), \mathbf{z}^L(2), \mathbf{z}^L(3), \ldots]_{q\times N}$ is the linear component on the output layer, this activation function must return a $q \times N$ matrix of probabilities. Hints: [2]

   - There are at least two ways in which this can be achieved.
   - `apply()`
   - `colSums()`

   (d) **Typeset mathematical and written response**: Let $\mathbf{X}^u$, and $\mathbf{X}^v$ denote the input **matrices** corresponding to the 'u' and 'v' sub-networks respectively and $\mathbf{Y}$ the matrix of responses. Then let $\mathbf{W}_1^u$ and $\mathbf{b}^{1u}$ denote the weight matrix and bias terms for the first hidden layer in the u-hemisphere, $\mathbf{W}_2^u$ and $\mathbf{b}^{2u}$ for the second hidden layer on the u-hemisphere and so on. Show how this network can be evaluated as a STANDARD $(2m, 2)$-network and give appropriate expressions/structures for the elements of the standard matrix-form updating equations. [3]

   (e) **Render code**: For purposes of modelling the present data, we'll place `AveSpeed` and `Sex` on the columns of $\mathbf{X}^u$ representing demographic variables, `Fluid` and `Carbs` on the columns of $\mathbf{X}^v$ representing nutrition variables, and encode the response for **two** outputs. Create these input and output matrices in R using the data set given. [1]

   (f) **Typeset mathematical and written response**: Let p^u and p^v denote the number of inputs on each respective hemisphere and q be the number of outputs. Write down expressions for the number of weights and biases respectively on a split-brain network with m nodes on the first hidden layer of each hemisphere. Also, evaluate the expressions for $m = 4$ for the present dataset. [1]

   (g) **Render code**: Write an R-function that takes two input matrices, `Xu` and `Xv`, and evaluates a forward pass of the split-brain network with m nodes on each [5]

of the hemispheres of the first hidden layer. Use tanh activations on all hidden nodes. This function must also include evaluation of an appropriate objective function and regularisation mechanism for the present dataset (let `nu` denote the regularization parameter and use the L2 norm).

Failover: If you cannot figure out how to evaluate this network you may use a standard (m, m)-network to conduct the analysis but you will be forgoing marks. In any case, see if you can get the analysis to work for the standard case first. That might be useful for saving time.

(h) **Render code + plot + in text response**: Conduct a validation analysis for $m = 4$ using standard R optimization routines in order to fit the models. Note: no scaling is required for the input variables here. Use `set.seed(2026)` before splitting the data into 80-20 training vs. validation. Plot the validation error vs. ν and use this figure to motivate your choice of regularization level. Indicate the level of regularisation chosen and report the value. `exp(seq(-5,3,length = n_val))` may be useful but experiment until you get the appropriate curve. [4]

(i) **R code + plots:** Draw a response curve over `Fluid`, `Carbs`, and `Sex` for a runner targeting 12km/h over the first half of the race. [3]

(j) **Typed response:** Interpret the response curves. (Try and answer the question at hand using this model. What advice would you give a runner targeting 12km/h over the first half of the race? And is there a differential effect for sex?) [2]

(k) Is there any empirical advantage to using the split-brain network for modelling the present data? Clearly motivate your response. [3]

(l) Three marks for the quality of your write-up. These are (some of) the things I'll be looking for: [3]

- Are your figures 1-1 aspect ratio?
- Are your figure captions full sentences and punctuated?
- Are your equations punctuated and properly typeset? Equations should read as sentences and thus should be punctuated.

Notes, Instructions, Etc.

Submission instructions:
For submission, I'd like you to give me

- A pdf containing your write-up. The write-up has an 8-page limit. Use the naming convention `STDNUM001_STDNUM002_Analytics_2026_A2.pdf` for your file. Note the underscores.
- Your R code. Use the naming convention `STDNUM001_STDNUM002_Analytics_2026_A2.R` for your code file. Your R code should NOT contain any of the following:

  ```
  install.packages()
  rm()
  setwd()
  ```

  I want to be able to run your code on my computer without having to manually edit your code, installing libraries or calling to external files.

ANY deviation from the above naming conventions WILL be penalised. Doing so wastes my time and I've grown quite weary of such simple instructions being ignored at my expense.